

# IoT Based Intelligent vehicle speed detection and Overspeed Alert System Using ESP32

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**ELECTRICAL AND ELECTRONICS ENGINEERING**

**Submitted by**

**B.Sahithi      23881A0206**

**R.Akshaya      23881A0243**

**B.Snehalatha   23881A0209**



**Department of Electrical and Electronics Engineering**



**Department of Electrical and Electronics Engineering**

## **CERTIFICATE**

This is to certify that the project titled **IoT Based Intelligent vehicle speed detection and Overspeed Alert System Using ESP32** is carried out by

**B.Sahithi      23881A0206**

**R.Akshaya      23881A0243**

**B.Snehalatha      23881A0209**

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Electrical And Electronics Engineering** during the year 2025-26.

# Abstract

Road safety remains a global concern as high speed vehicle movement contributes significantly to traffic accidents. This project presents the design and implementation of an automated speed monitoring system using the ESP32 microcontroller. The primary objective is to detect vehicle speed in real time and provide immediate local and remote alerts when a speed violation occurs. The system utilizes two Infrared (IR) sensors placed at a fixed distance to calculate the velocity of a passing vehicle based on the time interval between sensor triggers. A dual mode operation strategy is implemented to ensure system reliability; the device functions in Offline Mode for local alerts and Online Mode for cloud based notifications. When a vehicle exceeds the pre-defined speed limit, the system triggers a visual alert through a Red LED and an audible alarm via a Buzzer. Simultaneously, the speed data is displayed on a 16 by 2 I2C Liquid Crystal Display. If an active internet connection is detected, the ESP32 utilizes the Telegram Bot API to send an instant push notification to the authorities or the vehicle owner, providing the exact speed recorded. Conversely, if the speed is within the permissible limit, a Green LED is activated, and the status is displayed as safe. Experimental results demonstrate that the system provides high accuracy in speed measurement and reliable notification delivery. This low cost, scalable solution can be deployed in school zones, residential areas, and private parking facilities to enhance traffic rule enforcement and promote safer driving behavior.

**Keywords:** Internet of things (IoT), ESP32 Microcontroller, Overspeed detection, Infrared (IR) sensors, real-time monitoring, telegram bot notification

# Table of Contents

| Title                                                                                            | Page No. |
|--------------------------------------------------------------------------------------------------|----------|
| Abstract . . . . .                                                                               | i        |
| List of Tables . . . . .                                                                         | iv       |
| List of Figures . . . . .                                                                        | v        |
| Abbreviations . . . . .                                                                          | v        |
| <b>CHAPTER 1 Introduction</b> . . . . .                                                          | 1        |
| 1.1 Introduction . . . . .                                                                       | 1        |
| 1.2 Background and Motivation . . . . .                                                          | 1        |
| 1.3 Objectives of the Project Work . . . . .                                                     | 2        |
| 1.4 Organization of the Report . . . . .                                                         | 2        |
| <b>CHAPTER 2 Literature Survey</b> . . . . .                                                     | 3        |
| 2.1 Introduction . . . . .                                                                       | 3        |
| 2.2 Review of Prior Research . . . . .                                                           | 3        |
| 2.3 Identified Research Gaps . . . . .                                                           | 5        |
| 2.4 Summary . . . . .                                                                            | 5        |
| <b>CHAPTER 3 Prior Research on Vehicle Overspeed Detection</b> .                                 | 6        |
| 3.1 Introduction . . . . .                                                                       | 6        |
| 3.2 Existing Work . . . . .                                                                      | 6        |
| 3.3 Vehicle Speed Detection Techniques . . . . .                                                 | 7        |
| 3.4 Limitations of Existing System . . . . .                                                     | 8        |
| 3.5 Summary . . . . .                                                                            | 9        |
| <b>CHAPTER 4 Proposed Vehicle Overspeed Detection and Alert<br/>System Using ESP32</b> . . . . . | 10       |
| 4.1 Introduction . . . . .                                                                       | 10       |
| 4.2 Working Principle . . . . .                                                                  | 10       |
| 4.3 Hardware Implementation . . . . .                                                            | 11       |
| 4.4 Components Used . . . . .                                                                    | 12       |
| 4.4.1 ESP32 Microcontroller: . . . . .                                                           | 12       |
| 4.4.2 Infrared (Ir) Sensors: . . . . .                                                           | 12       |
| 4.4.3 16×2 LCD Display (I2C): . . . . .                                                          | 13       |
| 4.4.4 LED Indicators: . . . . .                                                                  | 13       |

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| 4.4.5 Buzzer: . . . . .                                           | 14        |
| 4.5 System Architecture . . . . .                                 | 14        |
| 4.6 Software Description . . . . .                                | 16        |
| 4.7 Advantages of Proposed System . . . . .                       | 22        |
| 4.8 Summary . . . . .                                             | 22        |
| <b>CHAPTER 5 Results and Discussion . . . . .</b>                 | <b>23</b> |
| 5.1 Introduction . . . . .                                        | 23        |
| 5.2 Experimental Setup . . . . .                                  | 23        |
| 5.3 Normal Operating Condition . . . . .                          | 24        |
| 5.4 Overspeed Condition . . . . .                                 | 24        |
| 5.5 IoT Monitoring Result . . . . .                               | 24        |
| 5.6 Evaluation of Quality Factors . . . . .                       | 25        |
| 5.7 Summary . . . . .                                             | 25        |
| <b>CHAPTER 6 Conclusions and Future Scope . . . . .</b>           | <b>26</b> |
| 6.1 Conclusions . . . . .                                         | 26        |
| 6.2 Future Scope of Work . . . . .                                | 27        |
| <b>Mapping of Program Outcomes (POs) and Sustainable Develop-</b> |           |
| <b>ment Goals (SDGs) . . . . .</b>                                | <b>28</b> |

## List of Tables

|     |                                           |    |
|-----|-------------------------------------------|----|
| 3.1 | Summary of Existing Systems . . . . .     | 7  |
| 3.2 | Vehicle Speed Conditions . . . . .        | 8  |
| 4.1 | Speed Limits Used for Detection . . . . . | 11 |

## List of Figures

|     |                                                              |    |
|-----|--------------------------------------------------------------|----|
| 4.1 | Block Diagram of Proposed Vehicle Overspeed Detection System | 11 |
| 4.2 | Block Diagram of Proposed System . . . . .                   | 12 |
| 4.3 | ESP32 Microcontroller . . . . .                              | 12 |
| 4.4 | IR Sensor . . . . .                                          | 13 |
| 4.5 | 16×2 LCD Display (I2C) . . . . .                             | 13 |
| 4.6 | LED Indicators . . . . .                                     | 14 |
| 4.7 | Buzzer . . . . .                                             | 14 |
| 4.8 | System Architecture . . . . .                                | 15 |
| 5.1 | Experimental Setup . . . . .                                 | 23 |
| 5.2 | Normal Condition . . . . .                                   | 24 |
| 5.3 | Overspeed Condition . . . . .                                | 24 |
| 5.4 | Telegram bot API Message . . . . .                           | 25 |

## Abbreviations

| Abbreviation | Description                                       |
|--------------|---------------------------------------------------|
| VCE          | Vardhaman College of Engineering                  |
| ESP32        | Microcontroller with built-in Wi-Fi and Bluetooth |
| IR           | Infrared Sensor                                   |
| IoT          | Internet of Things                                |
| LCD          | Liquid Crystal Display                            |
| LED          | Light Emitting Diode                              |
| Wi-Fi        | Wireless Fidelity                                 |
| API          | Application Programming Interface                 |
| I2C          | Inter-Integrated Circuit                          |
| GPIO         | General Purpose Input Output                      |

# CHAPTER 1

## Introduction

### 1.1 Introduction

Road accidents are a major global issue, with overspeeding being a leading cause of deaths and injuries. Many accidents occur when drivers exceed speed limits, especially in areas like school zones and residential regions. Traditional systems such as radar guns and surveillance cameras are expensive and complex to maintain. With the advancement of IoT, intelligent and cost-effective traffic monitoring systems can be developed. This project proposes an IoT-based vehicle overspeed detection system using the ESP32 microcontroller and IR sensors to measure speed in real time. The system operates in dual modes, providing local alerts through LEDs, a buzzer, and an LCD, and remote notifications via the Telegram Bot using Wi-Fi. The proposed system is low-cost, scalable, and helps improve traffic safety by promoting responsible driving behavior.

### 1.2 Background and Motivation

Rapid urbanization and the increasing number of vehicles have led to a rise in road accidents, with overspeeding being a major cause. Traditional speed monitoring systems are expensive and complex, limiting their widespread use. With the advancement of IoT technology, cost-effective and automated solutions can be developed. The ESP32 microcontroller enables real-time monitoring and communication. This project aims to design a low-cost, reliable, and scalable overspeed detection system that provides local alerts and remote notifications to improve road safety.

## 1.3 Objectives of the Project Work

This section outlines the specific goals that the project aims to achieve. The objectives are defined clearly to guide the design and develop an IoT-based vehicle overspeed detection and alert system using the ESP32 microcontroller. The system focuses on providing an efficient, real-time, and cost-effective solution for monitoring vehicle speed and ensuring road safety. It integrates sensing, processing, and communication technologies to detect overspeeding and generate alerts both locally and remotely.

The major objectives of the project work are listed below:

1. To develop a system for accurate vehicle speed measurement using IR sensors and ESP32.
2. To implement real-time local and remote alert mechanisms using LEDs, buzzer, LCD, and IoT.
3. To design a low-cost, scalable system with dual-mode (offline and online) operation.

## 1.4 Organization of the Report

The final section of Chapter 1 provides an outline of the structure of the entire report. It explains the organization and flow of the chapters and gives the reader an understanding of how the content will be presented. This section can be a simple list or paragraph summarizing each chapter's main points.

- Provide a brief description of what each chapter will cover.
- Typically, the structure would include:
  - **Chapter 2:** Literature Review
  - **Chapter 3:** Technical Title Related to the Prior Research
  - **Chapter 4:** Technical Title Related to the Proposed Work
  - **Chapter 5:** Results and Discussion
  - **Chapter 6:** Conclusions and Recommendations

# CHAPTER 2

## Literature Survey

### 2.1 Introduction

Vehicle overspeed detection has become an important research area in modern traffic management systems due to the rapid increase in vehicle usage and road accidents. Overspeeding is one of the major causes of accidents, leading to severe injuries, loss of life, and damage to property. Continuous monitoring of vehicle speed is essential to improve road safety and ensure proper traffic regulation [1].

Traditional speed monitoring systems such as radar guns and surveillance cameras are widely used but involve high installation and maintenance costs. Recent advancements in embedded systems and Internet of Things (IoT) technology have enabled the development of cost-effective and automated speed monitoring solutions using sensors and microcontrollers [2]. These systems allow real-time detection and remote alert generation.

This chapter presents a review of previous research related to vehicle speed detection methods, sensor-based systems, embedded platforms, and IoT-based monitoring techniques. The purpose of this study is to analyze existing approaches, identify their limitations, and highlight the need for a low-cost, reliable, and real-time overspeed detection system.

### 2.2 Review of Prior Research

Many researchers have studied vehicle speed detection systems and their role in improving road safety. Early research focused on traditional methods such as radar-based systems and surveillance cameras for detecting overspeeding and monitoring traffic conditions. These systems provide accurate results but involve high installation and maintenance costs [1].

Different sensor-based techniques have been developed using Infrared (IR)

sensors and microcontrollers for speed measurement. These systems calculate vehicle speed based on the time taken to travel between two fixed points. Such methods are simple and cost-effective but may be affected by environmental conditions [2]. Embedded systems have also been used for vehicle monitoring, including number plate recognition to identify vehicles involved in speed violations [3].

Surveillance-based approaches using image processing techniques have been introduced for vehicle detection and speed estimation. These systems use cameras and algorithms to track vehicle movement in real time. Although they provide reliable results, they require complex hardware and high computational resources [4]. Advanced systems aim to improve traffic monitoring using smart technologies and automated detection mechanisms [5].

Recent research focuses on computer vision and deep learning techniques for speed estimation. Video-based methods analyze vehicle motion using tracking algorithms and frame analysis [6]. Deep learning approaches using monocular cameras have achieved high accuracy by combining detection and tracking models [7]. Optimized vision-based systems have also been developed to improve efficiency and enable real-time performance on edge devices [8].

IoT-based monitoring systems have gained attention for enhancing road safety and traffic management. These systems enable real-time data transmission and alert generation through communication platforms such as Telegram and SMS [9]. IoT-based traffic monitoring systems also help in reducing congestion and improving traffic flow in smart cities [10].

Traditional methods such as inductive loop detectors are widely used for vehicle detection and traffic analysis [11]. Hybrid systems combining sensors, microcontrollers, and communication modules have also been proposed for real-time monitoring applications [12]. These systems provide better performance but may increase system complexity.

Therefore, there is a need for a low-cost, reliable, and efficient vehicle overspeed detection system that integrates both local alert mechanisms and remote monitoring capabilities to improve road safety.

## 2.3 Identified Research Gaps

From the review of previous research, it is observed that traditional vehicle speed detection systems such as radar and surveillance cameras are expensive and require complex infrastructure, limiting their widespread use. Sensor-based systems using IR sensors are cost-effective but may be affected by environmental conditions.

Many embedded and IoT-based systems lack integrated alert mechanisms or depend heavily on continuous internet connectivity, reducing their reliability. Additionally, some systems provide only local or remote alerts, but not both together.

Advanced approaches using vision and deep learning require high computational resources and complex setup. Therefore, there is a need for a low-cost, reliable system that provides accurate speed detection, real-time local alerts, remote monitoring, and dual-mode operation.

## 2.4 Summary

This chapter reviewed existing vehicle speed monitoring systems, including radar-based and IoT-enabled approaches. While traditional methods offer high accuracy, they are expensive and complex, whereas sensor-based and IoT systems are cost-effective but may have limitations in reliability and connectivity. The study identified key gaps such as lack of dual-mode functionality, dependency on internet connectivity, and limited scalability. Therefore, there is a need for a low-cost, reliable system with real-time detection and both local and remote alerts. The proposed system addresses these gaps using ESP32 and dual-mode operation, forming the basis for the following chapters.

## CHAPTER 3

### Prior Research on Vehicle Overspeed Detection

#### 3.1 Introduction

The purpose of this chapter is to present the prior research related to vehicle overspeed detection and alert system using the ESP32 microcontroller. Based on the identified research gaps, the system aims to provide a reliable, low-cost, and real-time monitoring solution by integrating both offline and online functionalities. It uses IR sensors to measure vehicle speed and the ESP32 to process the data and detect overspeeding. Alerts are generated through LEDs, a buzzer, and an LCD display, while IoT-based notifications are sent via the Telegram Bot API when internet connectivity is available. The following sections describe the system design, components, and working in detail.

#### 3.2 Existing Work

Existing vehicle speed detection systems have been developed using various technologies, including radar-based systems, camera-based surveillance, and sensor-based approaches. Radar and camera systems are widely used due to their high accuracy and reliability; however, they involve high installation and maintenance costs, making them less suitable for deployment in all areas.

In recent years, researchers have focused on low-cost alternatives using microcontrollers such as Arduino and ESP8266 along with Infrared (IR) sensors to measure vehicle speed. These systems calculate speed based on the time taken by a vehicle to pass between two sensors. Although these approaches are cost-effective and simple, they may face limitations due to environmental conditions and lack of advanced processing capabilities.

IoT-based systems have further enhanced speed monitoring by enabling real-time data transmission and remote alert mechanisms. Some systems

use cloud platforms or mobile applications to notify authorities about speed violations. However, many of these solutions depend heavily on continuous internet connectivity and do not provide reliable offline functionality.

Therefore, while existing works have contributed significantly to the field, there is still a need for a system that integrates accuracy, cost-effectiveness, and both offline and online alert capabilities, which is addressed in the proposed system.

**Table 3.1:** Summary of Existing Systems

| System               | Advantages          | Limitations               |
|----------------------|---------------------|---------------------------|
| Radar Systems        | High accuracy       | Expensive, complex setup  |
| Camera Systems       | Reliable monitoring | High cost, privacy issues |
| Sensor-Based Systems | Low cost, simple    | Affected by environment   |
| IoT-Based Systems    | Real-time alerts    | Needs internet connection |

### 3.3 Vehicle Speed Detection Techniques

Vehicle speed detection is an important aspect of traffic monitoring systems to ensure road safety and prevent accidents. In general, vehicle speed is measured based on the fundamental relationship between distance and time. In many practical systems, a predefined distance is maintained between two points, and the time taken by a vehicle to travel between these points is used to calculate its speed. Overspeeding is identified when the measured speed exceeds the permissible limit set for a particular road or zone.

Different techniques are used for vehicle speed detection. Traditional methods include radar-based systems, which use radio waves to measure vehicle speed with high accuracy. These systems are widely used in highways but require expensive equipment and complex installation. Camera-based systems use image processing techniques to detect vehicles and estimate speed, but they require high computational power and are affected by lighting conditions.

Sensor-based methods using Infrared (IR) sensors are widely used due to their simplicity and low cost. In this method, two IR sensors are placed at a fixed distance, and the time taken by a vehicle to pass between them is used to calculate speed. The speed is calculated using the formula:

$$v = \frac{d}{t}$$

where  $d$  is the distance between the sensors and  $t$  is the time taken.

**Table 3.2:** Vehicle Speed Conditions

| Condition | Speed Range  | Description      |
|-----------|--------------|------------------|
| Safe      | Within limit | Normal operation |
| Overspeed | Above limit  | Speed violation  |

Recent systems also use IoT technology to transmit speed data to cloud platforms or mobile applications for remote monitoring. These systems provide real-time alerts and improved accessibility. However, some existing methods depend heavily on internet connectivity and lack reliable offline operation. Therefore, there is a need for a system that provides accurate detection, real-time alerts, and both local and remote monitoring capabilities.

### 3.4 Limitations of Existing System

Despite significant advancements in vehicle speed detection systems, several limitations are observed in existing approaches. Traditional systems such as radar and camera-based methods are highly accurate but involve high installation and maintenance costs, making them unsuitable for widespread deployment. Sensor-based systems using IR sensors are cost-effective; however, their performance can be affected by environmental conditions such as lighting, dust, and weather.

Many IoT-based systems rely heavily on continuous internet connectivity for data transmission and alert generation, which limits their effectiveness in areas with poor network coverage. Vision-based systems using image processing and deep learning require high computational power and complex setup, increasing system complexity and cost. Additionally, some systems lack real-time response

or fail to integrate both local and remote alert mechanisms effectively.

These limitations highlight the need for a reliable, low-cost, and efficient system that can operate in both offline and online modes while ensuring accurate and real-time speed detection.

### 3.5 Summary

This chapter presented a comprehensive review of existing vehicle speed detection systems and related research works. Various approaches, including radar-based, sensor-based, IoT-based, and vision-based systems, were analyzed to understand their working principles and performance. Traditional systems provide high accuracy but are expensive and complex, whereas sensor-based and IoT-based systems offer cost-effective solutions with certain limitations. The review identified key challenges such as dependency on continuous internet connectivity, environmental sensitivity, and lack of integrated alert mechanisms. Additionally, many existing systems are not easily scalable for real-world deployment. Based on these findings, the need for a reliable, low-cost, and efficient system was established. The proposed system addresses these gaps by providing accurate speed detection along with both offline and online alert features. It ensures real-time monitoring and improved reliability. Thus, this chapter forms the basis for the design and implementation discussed in the subsequent chapters.

## CHAPTER 4

# Proposed Vehicle Overspeed Detection and Alert System Using ESP32

### 4.1 Introduction

This chapter presents the design and implementation of the proposed IoT-based vehicle overspeed detection and alert system using the ESP32 microcontroller. The system is developed to provide an efficient and low-cost solution for real-time vehicle speed monitoring.

Two Infrared (IR) sensors are used to measure the speed of a vehicle by calculating the time taken to travel between them. The ESP32 processes this data and compares it with a predefined speed limit. If the speed exceeds the limit, the system generates alerts through a red LED, buzzer, and LCD display, while a green LED indicates safe speed conditions.

In addition, the system supports online mode, where notifications are sent via the Telegram Bot API using Wi-Fi connectivity. This ensures both local and remote monitoring. The proposed system is reliable, scalable, and suitable for practical deployment in traffic-sensitive areas.

### 4.2 Working Principle

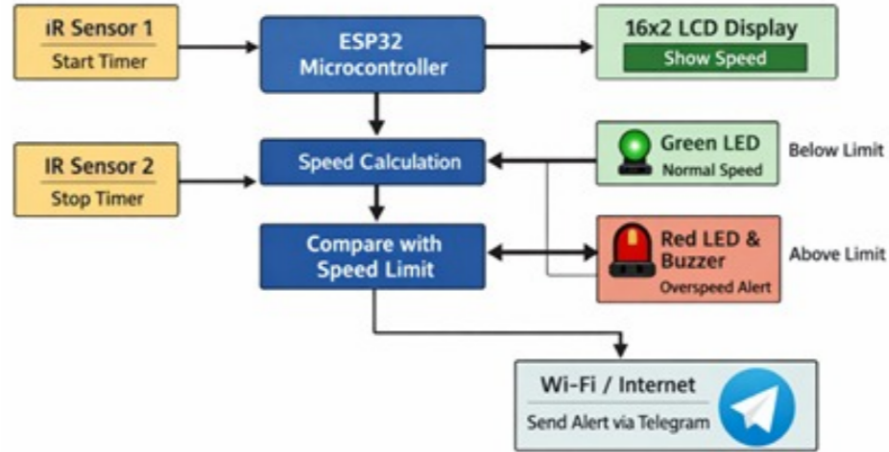
The vehicle movement is detected using two Infrared (IR) sensors placed at a fixed distance of 15 cm. When a vehicle crosses the first sensor, the ESP32 microcontroller records the start time, and when it crosses the second sensor, the end time is recorded. The time difference between these two events is calculated to determine the speed of the vehicle.

The speed is calculated using the following equation:

$$v = \frac{d}{t} \quad (4.1)$$

where  $d$  is the distance between the sensors and  $t$  is the time taken by the vehicle to travel between them.

Based on the calculated speed, a predefined limit is used to detect overspeed conditions. The ESP32 compares the measured speed with the set threshold value and generates alerts accordingly.



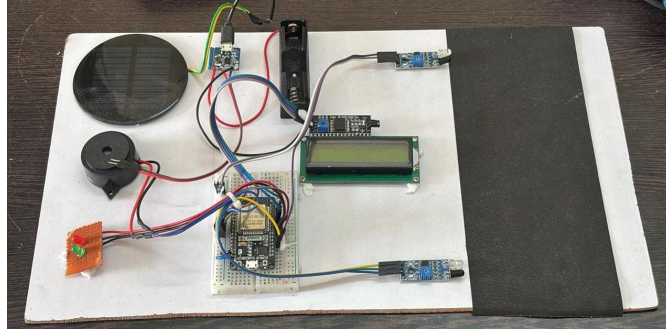
**Figure 4.1:** Block Diagram of Proposed Vehicle Overspeed Detection System

**Table 4.1:** Speed Limits Used for Detection

| Condition | Speed Range    | Description      |
|-----------|----------------|------------------|
| Safe      | $\leq 60$ cm/s | Normal operation |
| Overspeed | $> 60$ cm/s    | Speed violation  |

### 4.3 Hardware Implementation

The hardware implementation of the proposed system uses the ESP32 microcontroller as the main processing unit. Two IR sensors are placed at a fixed distance to detect vehicle movement and measure speed. A 16×2 LCD display is used to show speed and status, while LEDs and a buzzer provide alert indications. The system is powered using a battery or solar supply, and all components are connected using a breadboard and jumper wires. The ESP32's Wi-Fi module enables real-time notifications through the Telegram Bot API.

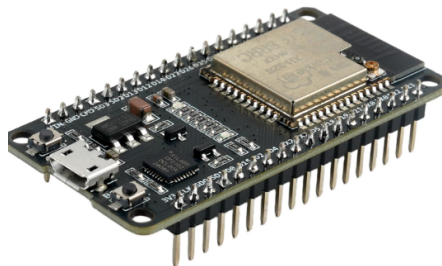


**Figure 4.2:** Block Diagram of Proposed System

## 4.4 Components Used

### 4.4.1 ESP32 Microcontroller:

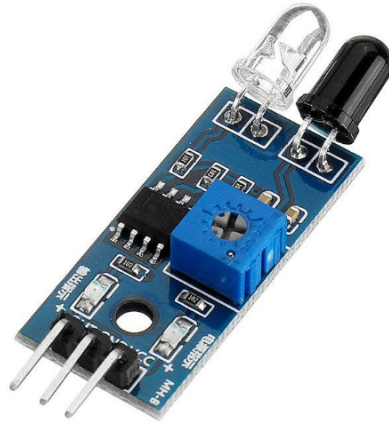
The ESP32 is a powerful microcontroller with a dual-core processor operating up to 240 MHz. It includes built-in Wi-Fi and Bluetooth capabilities, making it suitable for IoT applications. In this system, it acts as the central controller, processing input signals from IR sensors, calculating vehicle speed, and controlling output devices such as LEDs, buzzer, and LCD. It also handles communication with the Telegram Bot API for real-time alerts.



**Figure 4.3:** ESP32 Microcontroller

### 4.4.2 Infrared (Ir) Sensors:

IR sensors consist of an IR transmitter and receiver pair. The transmitter emits infrared light, and the receiver detects reflected light when an object passes. In this project, two IR sensors are placed at a fixed distance to detect vehicle movement. The time difference between sensor activations is used to calculate speed. These sensors are low-cost, fast in response, and suitable for real-time detection.



**Figure 4.4:** IR Sensor

#### 4.4.3 16×2 LCD Display (I2C):

The 16×2 LCD display is used to show the vehicle speed and system status. It can display 16 characters per line across two lines. The I2C interface reduces the number of pins required from 6–8 to just 2 (SDA and SCL), simplifying wiring and improving system efficiency.



**Figure 4.5:** 16×2 LCD Display (I2C)

#### 4.4.4 LED Indicators:

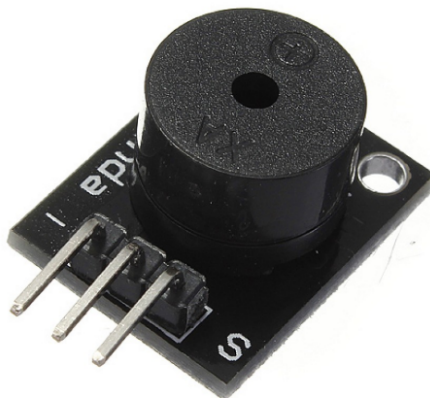
LEDs are used as visual indicators in the system. The green LED indicates that the vehicle is within the safe speed limit, while the red LED indicates overspeeding. LEDs are energy-efficient, have a long lifespan, and provide instant visual feedback.



**Figure 4.6:** LED Indicators

#### **4.4.5 Buzzer:**

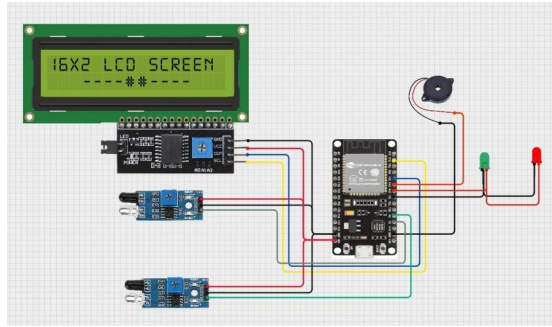
The buzzer is used to provide an audible alert when overspeeding is detected. An active buzzer is typically used, which produces sound when powered. It helps in attracting immediate attention and enhances safety by providing sound-based warnings.



**Figure 4.7:** Buzzer

### **4.5 System Architecture**

The system architecture of the proposed vehicle overspeed detection and alert system is designed to ensure accurate speed measurement, real-time processing, and effective alert mechanisms. The system is broadly divided into three main units: input unit, processing unit, and output unit.



**Figure 4.8:** System Architecture

The input unit consists of two Infrared (IR) sensors placed at a fixed distance along the path of the vehicle. These sensors detect the presence of a moving vehicle and generate signals when the vehicle passes each sensor. The time interval between the activation of the two sensors is recorded and used to calculate the speed of the vehicle.

The processing unit is built around the ESP32 microcontroller, which acts as the core component of the system. It receives signals from the IR sensors, computes the time difference, and calculates the vehicle speed using a predefined formula. The calculated speed is then compared with a predefined speed limit to determine whether the vehicle is overspeeding.

The output unit includes a red LED, green LED, buzzer, and a 16×2 I2C LCD display. If the vehicle exceeds the speed limit, the system activates the red LED and buzzer to provide immediate local alerts, and the speed value is displayed on the LCD. If the vehicle is within the safe limit, the green LED is activated to indicate normal conditions.

In addition to local alerts, the system supports an online communication module using the ESP32's built-in Wi-Fi capability. When connected to the internet, the system sends real-time notifications through the Telegram Bot API, including the detected speed and alert status. This dual-mode operation ensures that the system remains functional even in the absence of internet connectivity.

Overall, the architecture is simple, cost-effective, and scalable, making it suitable for deployment in various traffic monitoring scenarios such as school zones, residential areas, and parking facilities.

## 4.6 Software Description

The software for the proposed system is developed using the Arduino IDE and programmed in Embedded C/C++. The ESP32 reads signals from IR sensors, calculates the time interval, and determines vehicle speed. Based on the speed, it controls outputs such as LEDs, buzzer, and LCD display to indicate safe or overspeed conditions.

The system also uses the ESP32's Wi-Fi to send real-time alerts through the Telegram Bot API. A non-blocking approach ensures continuous monitoring and reliable operation in both offline and online modes.

```
1 #include <WiFi.h>
2 #include <WiFiClientSecure.h>
3 #include <UniversalTelegramBot.h>
4 #include <ArduinoJson.h>
5 #include <LiquidCrystal_I2C.h>
6
7 // ——— Credentials ———
8 const char* ssid = "YOUR_WIFI_NAME";
9 const char* password = "YOUR_WIFI_PASSWORD";
10
11 #define BOTtoken "YOUR_TELEGRAMBOT_TOKEN"
12 #define CHAT_ID "YOUR_CHAT_ID"
13
14 // ——— Hardware Pins ———
15 #define SENSOR_ENTRY 2
16 #define SENSOR_EXIT 4
17 #define LED_GREEN 5
18 #define LED_RED 18
19 #define BUZZER 19
20
21 // ——— Settings ———
22 float sensorDistance = 15.0; // Distance between sensors
```

```

        in cm
23 float speedLimit = 60.0;           // Speed limit in cm/s
24
25 LiquidCrystal_I2C lcd(0x27, 16, 2);
26
27 WiFiClientSecure client;
28 UniversalTelegramBot bot(BOTtoken, client);
29
30 void setup() {
31     Serial.begin(115200);
32
33     lcd.init();
34     lcd.backlight();
35
36     pinMode(SENSOR_ENTRY, INPUT);
37     pinMode(SENSOR_EXIT, INPUT);
38
39     pinMode(LED_GREEN, OUTPUT);
40     pinMode(LED_RED, OUTPUT);
41     pinMode(BUZZER, OUTPUT);
42
43     lcd.print("Init WiFi...");
44
45     WiFi.begin(ssid, password);
46
47     client.setCACert(TELEGRAM_CERTIFICATE_ROOT);
48
49     // Wait maximum 5 seconds for Wi-Fi
50     unsigned long startAttemptTime = millis();
51
52     while (WiFi.status() != WL_CONNECTED &&
53           millis() - startAttemptTime < 5000) {

```

```

54     delay(500);
55     Serial.print(".");
56 }
57
58 lcd.clear();
59
60 if (WiFi.status() == WL_CONNECTED) {
61     lcd.print("Mode: ONLINE");
62     bot.sendMessage(CHAT_ID,
63                     "System Online – Vehicle Speed
64                     Monitoring",
65                     "");
66 }
67 else {
68     lcd.print("Mode: OFFLINE");
69 }
70
71 delay(2000);
72
73 lcd.clear();
74 lcd.print("Scanning...");
75 }
76
77 void loop() {
78     // Detect vehicle at first sensor
79     if (digitalRead(SENSOR_ENTRY) == LOW) {
80
81         unsigned long startTime = millis();
82
83         // Wait for second sensor
84         while (digitalRead(SENSOR_EXIT) == HIGH) {

```

```

85
86         // Safety timeout
87         if (millis() - startTime > 2000) {
88             break;
89         }
90     }
91
92     unsigned long endTime = millis();
93
94     float timeTaken =
95         (endTime - startTime) / 1000.0;
96
97     float speed =
98         (timeTaken > 0) ?
99         (sensorDistance / timeTaken) : 0;
100
101     processSpeed(speed);
102 }
103 }
104
105 void processSpeed(float speed) {
106
107     lcd.clear();
108
109     lcd.print("Speed: ");
110     lcd.print(speed);
111     lcd.print(" cm/s");
112
113     if (speed > speedLimit) {
114
115         // Local overspeed alert
116         digitalWrite(LED_RED, HIGH);

```

```

117     digitalWrite(BUZZER, HIGH);
118
119     lcd.setCursor(0, 1);
120     lcd.print("OVERSPEED!");
121
122     // IoT alert through Telegram
123     if (WiFi.status() == WL_CONNECTED) {
124
125         String message =
126             "Overspeed Alert!\nSpeed: " +
127             String(speed) +
128             " cm/s";
129
130         bot.sendMessage(CHAT_ID, message, "");
131     }
132
133 }
134 else {
135
136     // Safe speed indication
137     digitalWrite(LED_GREEN, HIGH);
138
139     lcd.setCursor(0, 1);
140     lcd.print("SAFE SPEED");
141 }
142
143 delay(3000);
144
145 resetSystem();
146 }
147
148 void resetSystem() {

```

```
149
150     digitalWrite(LED_RED, LOW);
151     digitalWrite(LED_GREEN, LOW);
152     digitalWrite(BUZZER, LOW);
153
154     lcd.clear();
155     lcd.print(" Scanning ...");
156 }
```

**Listing 4.1:** ESP32 Vehicle Overspeed Detection Code

## 4.7 Advantages of Proposed System

- Low-cost implementation using simple and easily available components.
- Provides accurate real-time vehicle speed measurement using IR sensors.
- Dual-mode operation (offline and online) ensures reliable performance.
- Generates immediate local alerts using LEDs, buzzer, and LCD display.
- Supports remote monitoring through IoT using the Telegram Bot API.
- Easy to install, maintain, and deploy in various locations.
- Scalable and suitable for applications such as school zones and residential areas.
- Energy-efficient operation with support for battery and solar power.

## 4.8 Summary

This chapter presented the design and implementation of the proposed vehicle overspeed detection and alert system using the ESP32 microcontroller. The system architecture, hardware components, and software design were explained in detail. The working principle based on IR sensors and time-based speed calculation was also discussed. The integration of output devices such as LEDs, buzzer, and LCD for local alerts, along with IoT-based notifications using the Telegram Bot API, was described. Overall, the chapter demonstrated how the proposed system provides an efficient, low-cost, and reliable solution for real-time vehicle speed monitoring with both offline and online functionalities.

# CHAPTER 5

## Results and Discussion

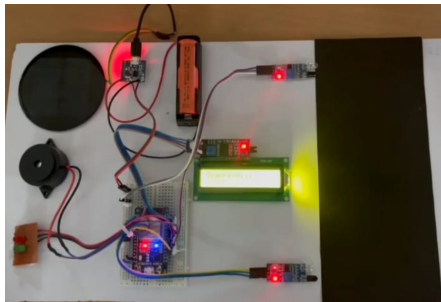
### 5.1 Introduction

This chapter presents the results obtained from the implementation of the proposed vehicle overspeed detection and alert system and discusses their significance. The results demonstrate the system's ability to accurately measure vehicle speed and generate appropriate alerts under different conditions. The performance of both offline and online modes is evaluated to ensure reliability and effectiveness.

This chapter is organized to first present the experimental results, followed by a detailed discussion of system performance, accuracy, and limitations. The analysis highlights how the proposed system meets the project objectives and contributes to improving real-time traffic monitoring and safety.

### 5.2 Experimental Setup

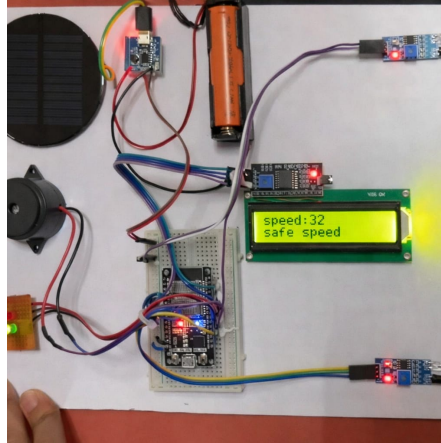
The hardware prototype of the proposed system was developed using the ESP32 microcontroller, IR sensors, LCD display, LEDs, buzzer, and Wi-Fi module. The system was tested by allowing objects to pass across the sensors to simulate vehicle movement. The speed was calculated based on the time taken between the sensors, and the response of the system was observed.



**Figure 5.1:** Experimental Setup

### 5.3 Normal Operating Condition

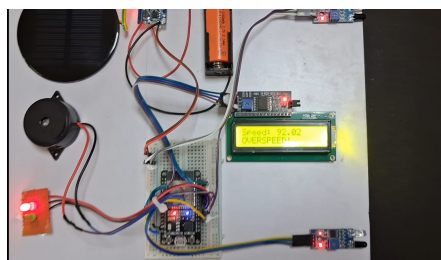
The system was first tested under normal speed conditions. The measured speed was within the predefined safe limit. The system displayed a "SAFE" status on the LCD and activated the green LED. No alerts were generated.



**Figure 5.2:** Normal Condition

### 5.4 Overspeed Condition

Overspeed condition was created by reducing the time between sensor triggers. The system detected the overspeed correctly and activated the red LED and buzzer. The LCD displayed an "OVERSPEED" warning message.



**Figure 5.3:** Overspeed Condition

### 5.5 IoT Monitoring Result

The system successfully transmitted speed data using Wi-Fi. Alerts were sent to the user through the Telegram Bot API when overspeed was detected. This enabled real-time remote monitoring of vehicle speed.

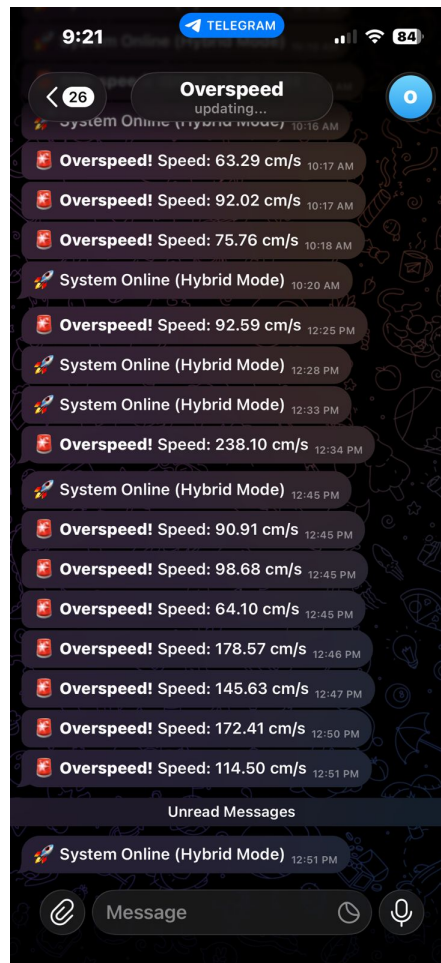


Figure 5.4: Telegram bot API Message

## 5.6 Evaluation of Quality Factors

**Environment:** Helps reduce accidents and improves road safety.

**Sustainability:** Low-cost and energy-efficient system.

**Safety:** Provides immediate alerts for overspeed conditions.

**Ethics:** Promotes responsible driving behavior.

**Cost:** Economical compared to traditional systems.

**Type:** Suitable for small-scale and real-time applications.

## 5.7 Summary

The proposed system successfully detected vehicle speed and identified overspeed conditions. It is efficient, low-cost, real-time traffic monitoring.

## CHAPTER 6

### Conclusions and Future Scope

#### 6.1 Conclusions

This project presents the design and implementation of an IoT-based vehicle overspeed detection and alert system using the ESP32 microcontroller. The system successfully measures vehicle speed using IR sensors and detects overspeed conditions with good accuracy in real time. It provides immediate local alerts through LEDs, a buzzer, and an LCD display, ensuring a quick response to speed violations. In addition, the integration of Wi-Fi and the Telegram Bot API enables remote monitoring and instant notification to users or authorities. The system operates efficiently in both offline and online modes, ensuring reliable performance even in the absence of internet connectivity. The experimental results confirm the effectiveness, accuracy, and responsiveness of the system under different conditions. Furthermore, the system is energy-efficient and can be powered using battery or solar sources, making it suitable for outdoor deployment. The proposed solution is low-cost, easy to implement, scalable, and adaptable for various environments such as school zones, residential areas, highways, and parking facilities. It can also be integrated with smart city infrastructure for improved traffic management. Overall, the project achieves its objectives and provides a practical solution for real-time speed monitoring, contributing significantly to road safety and promoting responsible driving behavior.

## 6.2 Future Scope of Work

The proposed vehicle overspeed detection system can be further enhanced in several ways to improve its performance and applicability. The current system can be extended by integrating GPS modules to track the exact location of overspeeding vehicles, which would help authorities take more effective action. The use of advanced sensors or camera-based systems can improve accuracy and enable features such as number plate recognition.

Future work may also include the development of a mobile application or web dashboard for better visualization and management of speed data. Cloud integration can be enhanced to store and analyze historical data for traffic pattern analysis. Additionally, the system can be scaled for large-scale deployment on highways and smart city applications.

Improvements in power management, such as optimized solar charging systems, can make the system more energy-efficient for outdoor use. Further research can focus on improving reliability under different environmental conditions and reducing false detections. Overall, these enhancements can make the system more robust, intelligent, and suitable for real-world traffic monitoring and safety applications.

# Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

## 1. Contribution to Program Outcomes (POs) and Program Specific Outcomes (PSOs)

The project contributes to the attainment of the following Program Outcomes:

### **PO1: Engineering Knowledge**

This project demonstrates the application of fundamental knowledge in electronics, embedded systems, and IoT. The design, analysis, and implementation phases required the use of core principles related to microcontrollers, sensors, and wireless communication.

### **PO2: Problem Analysis**

The project involves problem identification and systematic analysis of vehicle overspeeding, which is a major cause of road accidents. Appropriate tools and techniques were used to analyze sensor data and develop an effective monitoring system.

### **PO3: Design/Development of Solutions**

The system developed in this project addresses real-world challenges such as traffic safety and overspeed detection. The proposed solution meets specified needs with consideration of constraints like cost, reliability, and real-time operation.

### **PO4: Conduct Investigation of Problems**

Experimental testing was carried out to verify the performance of the system under different speed conditions. The accuracy of speed measurement and alert generation was analyzed to ensure reliable operation.

### **PO5: Modern Tool Usage**

The project uses modern tools such as Arduino IDE, ESP32 microcontroller,

IR sensors, and IoT platforms like Telegram for real-time monitoring and alert notification.

#### **PO6: The Engineer and Society**

The developed system improves road safety by detecting overspeeding and providing timely alerts, which is beneficial for public safety in areas such as schools, residential zones, and highways.

#### **PO7: Environment and Sustainability**

The system supports sustainable operation by using low-power components and can be powered using battery or solar energy, reducing energy consumption.

#### **PO8: Ethics**

The project follows engineering ethics by developing a reliable and safe system that promotes responsible driving behavior and ensures public safety.

#### **PO9: Individual and Team Work**

The project was completed through effective teamwork involving hardware design, software development, testing, and documentation.

#### **PO10: Communication**

Technical reports, presentations, and documentation were prepared to clearly explain the system design, implementation, and results.

#### **PO11: Project Management and Finance**

The project was completed within a limited budget using low-cost components such as ESP32, IR sensors, and basic electronic modules.

#### **PO12: Life Long Learning**

The project helped in learning new technologies such as IoT, embedded systems, wireless communication, and real-time monitoring applications.

### **2. Program Specific Outcomes (PSOs)**

#### **PSO1: Apply the knowledge of domain-specific skill set for the design and analysis of components in VLSI and Embedded systems.**

The project applies domain-specific knowledge in embedded systems and IoT

for the design and analysis of a vehicle overspeed detection and alert system using the ESP32 microcontroller, sensors, and communication modules.

**PSO2: Demonstrate the technical competency and use appropriate techniques in the realization of advanced communication systems.**

The project demonstrates technical competency by using appropriate tools and techniques such as wireless communication, IoT platforms, and real-time data transmission through the Telegram Bot API.

**3. Alignment with Sustainable Development Goals (SDGs)**

The project supports the following Sustainable Development Goals:

**SDG 11: Sustainable Cities and Communities**

The project contributes to this goal by improving road safety through real-time vehicle speed monitoring and alert systems, helping to reduce accidents and promote safer transportation systems.

**SDG 9: Industry, Innovation and Infrastructure**

The project contributes to this goal by developing an innovative IoT-based system using modern technologies like ESP32 and wireless communication for smart traffic monitoring.

**Overall Impact**

The project not only fulfills academic learning outcomes but also contributes toward broader societal and environmental objectives. By integrating technical knowledge with sustainability considerations, the work demonstrates alignment between institutional Program Outcomes and global development priorities.

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